



Deficit Irrigation of Alfalfa for Water-Savings in the Great Plains and Intermountain West: A Review and Analysis of the Literature

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ABSTRACT

Diversions of water from irrigated agriculture are occurring in the western United States to address increasing municipal and industrial demands. Deficit irrigation of alfalfa (*Medicago sativa* L.) could be a source of water without complete dry-up of irrigated fields. Water saving potential from alfalfa is high because it is a high water-use crop produced on 12% of the irrigated land in the United States. The objectives of this paper are to review alfalfa plant–water relations in the Great Plains and Intermountain West, to understand potential water savings through deficit irrigation, and to identify management practices that maximize water-use efficiency (WUE). Alfalfa biomass yield exhibits a linear relationship to evapotranspiration (ET) with the slope of a regionally aggregated water production function of $0.16 \text{ Mg ha}^{-1} \text{ cm}^{-1}$. Relative ET declines 30% faster than relative biomass yield under deficit irrigation or dryland management. Because early season harvests have greater WUE, combining full irrigation in spring with no irrigation during less efficient water-use growth periods may be more effective in saving water than season-long deficit irrigation. Management practices that can influence WUE under deficit irrigation include stand age, growth stage at harvest, and alfalfa variety. A potential complication with controlled deficit irrigation of alfalfa is an uncertain contribution to ET from a water table. As alfalfa roots develop over time, a significant percentage of total ET can come from water tables shallower than 200 cm and the percentage increases as availability of water from precipitation or irrigation declines.

IN WATER-SCARCE ENVIRONMENTS, demand for water exceeds the available supply, creating competition among users for this limited resource. Population growth in the western United States drives water scarcity and the problem is magnified by drought conditions and declining groundwater levels (McGuire, 2004; Dugan and Sharpe, 1996). As an example, the Statewide Water Supply Initiative compiled by the Colorado Water Conservation Board (2004a, 2004b) projects that Colorado's population will grow by 65% within the next 25 yr resulting in a likely dry-up of 170,000 ha of irrigated farmland to meet the urban demand for water. Similar water reallocation from agriculture to municipalities has the potential to dry up irrigated agriculture throughout much of the western United States. Solutions that meet water demands of growing populations and industry without loss of irrigated land are needed to preserve food production and regional economies that depend on irrigated agriculture.

Because alfalfa is a high water-use crop (Stanberry, 1955; Schneekloth and Andales, 2009) grown on 12% of the irrigated land in the United States (National Agricultural Statistics Service, 2007), controlled deficit irrigation of alfalfa has been proposed as a source of water transfers from agriculture (Putnam

et al., 2005). Controlled deficit irrigation is an approach that supplies water at a rate below the full crop water requirement and usually results in lower biomass yields than full irrigation (Carter and Sheaffer, 1983; Grimes et al., 1992; Undersander 1987). Established alfalfa may be adapted to deficit irrigation with drought avoidance mechanisms such as deep rooting (Peterson, 1972) and drought induced dormancy (Peterson, 1972; Smith, 1962; Robinson and Massengale, 1968).

The objectives of this paper were to review alfalfa plant–water relations in the Great Plains and Intermountain West regions of the United States in an effort to identify practices to maximize WUE under deficit irrigation. Irrigated alfalfa studies were compared to evaluate differences in ET, biomass yield, WUE, and the potential for water savings from controlled deficit irrigation.

Evapotranspiration, Biomass Yield, and Water-Use Efficiency under Variable Irrigation

Studies conducted in the Great Plains and Intermountain West of the United States were compared to evaluate alfalfa ET and biomass yield under variable irrigation (Daigger et al., 1970; Bauder et al., 1978; Retta and Hanks, 1980; Sammis, 1981; Carter and Sheaffer, 1983; Undersander, 1987; Wright, 1988; Bogler and Matches, 1990; Smeal et al., 1991). Only literature sources that had a complete accounting of water use from all sources, including irrigation, precipitation, and soil water, were included in this comparison. The majority of these studies used a water balance approach to determine ET. Studies that did not account for soil water changes or that may have been confounded by uncontrolled water use from a water table were excluded. Our analysis revealed that growing season ET averaged 91 cm over full irrigation treatments, 80 cm for deficit

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Abbreviations: ET, evapotranspiration; WUE, water-use efficiency; Gc, cumulative growing-degree-days; Sd, average daily solar irradiance

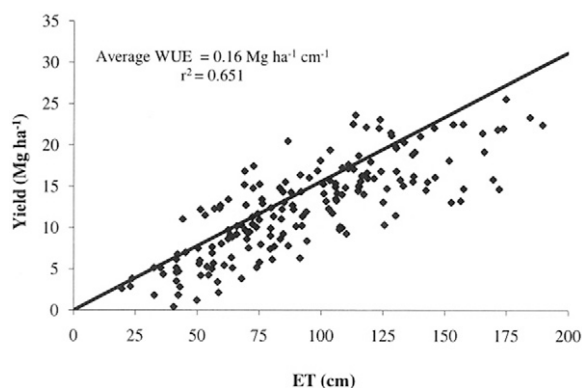


Fig. 1. Season long alfalfa biomass yield and evapotranspiration (ET) from studies with variable irrigation in the Great Plains and Intermountain West of the United States. Each study (not each point) was weighted equally to create the regression line. The slope of the line represents a regional average water-use efficiency (WUE) of $0.16 \text{ Mg ha}^{-1} \text{ cm}^{-1}$.

irrigation treatments, and 39 cm for dryland treatments (Table 1). Average annual biomass yield was 16.6 Mg ha^{-1} under full irrigation, 11.1 Mg ha^{-1} under a variety of deficit irrigation treatments, and 6.0 Mg ha^{-1} under dryland conditions. Although the irrigation treatments are pooled over a wide range of climatic conditions, the comparison illustrates the magnitude of water use by irrigated alfalfa, the potential water savings from converting full irrigation alfalfa to deficit irrigation or dryland, and the extent of biomass yield reduction that would accompany the reduced irrigation levels.

A regional average WUE for alfalfa was estimated from the slope of a water production function created by aggregating the results of the reviewed studies. To illustrate the spatial and temporal variability in the water production function, results from individual study years were used (Fig. 1) rather than multi-year treatment averages (Table 1). Alfalfa biomass yield responds in a positive, linear relationship to increasing ET, with an average WUE of $0.16 \text{ Mg ha}^{-1} \text{ cm}^{-1}$. This WUE is consistent with values reported for this region by others (Hanson and Putnam, 2000). There is wide variation in reported WUE, with total seasonal WUE values ranging from 0.08 to $0.23 \text{ Mg ha}^{-1} \text{ cm}^{-1}$ (Table 1 and Fig. 1). To normalize ET and biomass yield for different climatic and environmental conditions, a relative water production function with multi-year average data was created (Fig. 2). Relative biomass yield declines at a greater rate than relative ET (slope = 1.28), indicating that WUE will decrease with decreasing ET under either deficit irrigation or dryland conditions. The relative water production function (Fig. 2) can be used with localized estimates of maximum yield and ET to predict alfalfa production under planned reductions in irrigation.

Several individual studies confirm decreasing WUE with deficit irrigation over a wide range of environmental conditions. Retta and Hanks (1980) used line source irrigation in an alfalfa study in Utah and found that WUE ranged from $0.19 \text{ Mg ha}^{-1} \text{ cm}^{-1}$ under deficit irrigation to $0.21 \text{ Mg ha}^{-1} \text{ cm}^{-1}$ for full irrigation (Table 1). Sammis (1981) found that WUE ranged from $0.09 \text{ Mg ha}^{-1} \text{ cm}^{-1}$ in the driest conditions to $0.15 \text{ Mg ha}^{-1} \text{ cm}^{-1}$ in the wettest conditions under line source irrigation in New Mexico (Table 1). Another New Mexico study under line source irrigation (Smeal et al., 1991) supports these findings; WUE ranged from $0.07 \text{ Mg ha}^{-1} \text{ cm}^{-1}$ in the driest conditions

to $0.14 \text{ Mg ha}^{-1} \text{ cm}^{-1}$ in the wettest conditions (Fig. 3). A similar observation of declining WUE with ET was observed in a more humid environment in Minnesota, where average WUE values were 0.23, 0.23, 0.21, and $0.12 \text{ Mg ha}^{-1} \text{ cm}^{-1}$ for high, medium high, medium low, and rainfed irrigation treatments (Carter and Sheaffer, 1983). Thus, over a wide range of environmental conditions, there is a consistent trend of decreased WUE with decreasing ET under deficit irrigation.

Seasonal Differences in Water-Use Efficiency

Water-use efficiency varies with harvest interval due to differences in environmental conditions and plant carbohydrate partitioning. Smeal et al. (1991) compared biomass yield and transpiration over harvest interval. Greatest WUE was measured during the first harvest of each season, and the remaining harvests had lesser WUE (Table 2). A study conducted by Underlander (1987) in Texas also found that the greatest WUE was observed for the first harvest (Table 2), and the trend is echoed in the results of both Daigger et al. (1970) and Wright (1988).

Variation in solar irradiance helps explain seasonal differences in WUE. Smeal et al. (1991) compared biomass yield per unit of transpiration with level of solar irradiance over time. The results of this experiment showed that biomass yield increased with increasing average daily solar irradiance (S_d) during the growing period from a low of $16.1 \text{ MJ m}^{-2} \text{ d}^{-1}$ to a maximum of $28.0 \text{ MJ m}^{-2} \text{ d}^{-1}$. Values of S_d greater than $28 \text{ MJ m}^{-2} \text{ d}^{-1}$ resulted in biomass yields that decreased slowly or plateaued. Based on a plant growth simulation model, the increase in biomass yield per unit of transpiration was explained by increasing S_d due to increased light penetration into the canopy rather than an increase in heat energy (Holt et al., 1975). Both conditions, high light and relatively low temperatures only occur in the spring, whereas, light levels are limiting in the fall. Thus, harvest intervals corresponding to the greatest WUE occur when solar irradiance is high enough to induce high levels of photosynthesis and temperatures are low enough to keep evaporation at a minimum, such as the first harvest in the spring (Delaney et al., 1974; Leavitt et al., 1979).

The seasonal variation in WUE also relates to carbohydrate reserve flux in the alfalfa plant. Biomass development early in the season depends on carbohydrate reserves accumulated during the fall of the previous season and results in the highest WUE of the season (Smith, 1962; Robinson and Massengale, 1968). After the first harvest, carbohydrates for growth and restoration of root reserves come from photosynthesis in new leaves, and transpiration is correlated with leaf surface area (MacAdam and Barta, 2007). As day-length shortens and temperatures decline in the late summer and early fall, greater amounts of photosynthate are partitioned into root reserves, resulting in lower levels of above-ground biomass, yield, and WUE (Hanson et al., 1988).

These observations suggest an approach to saving water from irrigated alfalfa may be to concentrate irrigation during periods of the growing season that have the greatest WUE, such as the first harvest interval, followed by deficit or no irrigation during periods of the growing season with the least WUE, such as mid-summer harvest intervals. This approach, referred to as irrigation termination or partial season irrigation, has the potential to improve WUE relative to a controlled deficit irrigation that is applied uniformly through the growing season. None of the studies reported in Table 1 managed irrigation in this manner,

but results from other regions show improved WUE with this irrigation method. Putnam et al. (2005) showed significant water savings from partial season irrigation of alfalfa in the Klamath Basin and Sacramento Valley of California with few long-term impacts on alfalfa stands. However, for arid climates and sandy soils, summer termination of irrigation can reduce alfalfa stands and biomass yields after irrigation is resumed (Ottman et al., 1996). More information is needed about partial season irrigation in the Great Plains and Intermountain West regions.

Management Factors Affecting Alfalfa Water-Use Efficiency

Variety

While it is expected that alfalfa varieties would vary widely in WUE, there is little evidence to support this. A Utah line-source irrigation study evaluated WUE of alfalfa varieties 'Ladak', 'Washoe', and 'Mesilla' with fall dormancy ratings of three, five, and seven, respectively (Retta and Hanks, 1980). The 2-yr study found no difference in biomass yield or water use among varieties. A line-source irrigation study in Texas (Undersander, 1987) examined WUE of alfalfa varieties 'Vanguard', 'Cody', 'Zia', and 'Dawson', all with varying dormancy characteristics and diverse genetic backgrounds. Results showed no significant differences in WUE among the varieties for any level of irrigation. Hattendorf et al. (1990) conducted a line-source irrigation study in Washington comparing water use and biomass yield of alfalfa varieties 'Vernal', 'Vernema', and 'CUF 101' with fall dormancy ratings of two, four, and nine, respectively. No differences in biomass yield or water use were found. A line-source irrigation study conducted in California (Grimes et al., 1992) evaluated WUE of alfalfa varieties 'CUF 101', 'Moapa 69', and 'WL 318' with fall dormancy ratings of 9, 8, and 3, respectively. In the California environment, the semi-dormant 'WL 318' had a slightly higher WUE than the other two varieties during the cool spring conditions. During the hot summer conditions, the less dormant CUF 101 and Moapa 69 varieties had higher WUE. However, there was no significant difference among varieties when WUE was evaluated on a total season basis. These studies indicated that alfalfa varieties with varying dormancy traits and genetic backgrounds did not differ greatly in total-season WUE. Varieties with different fall dormancy may not relate to the suitability of the variety under deficit irrigation.

Less is known about the potential role of alfalfa variety on partial season irrigation. It has been reported that alfalfa varieties that have cold tolerance and winter hardiness also have drought tolerance because both conditions desiccate plant cells (Jung and Larson, 1972). Drought and cold tolerance may be linked to small cell size. If advantages in WUE during particular periods of the growing season do exist, selecting varieties that possess drought tolerance and are more dormant may be best for stand survival during dry periods in a partial season irrigation system. More studies are needed to evaluate alfalfa varieties under partial season irrigation in different environments. It is thought that a more fibrous root system may be more efficient in extracting soil water than a dominant tap root system. However, it should be noted that numerically high fall dormancy ratings tend to be correlated with tap-rooted alfalfa varieties and low fall dormancy ratings are correlated with fibrous-rooted alfalfa varieties (Smith, 1993).

Table 1. Average total season biomass yield, evapotranspiration (ET), and water-use efficiency (WUE) from studies of alfalfa under variable irrigation in the Great Plains and Intermountain West of the United States.

Author	Location	Treatment	Yield Mg ha ⁻¹	ET cm	WUE Mg ha ⁻¹ cm ⁻¹
Daigger et al., 1970	NE	full irrigation	11.5	151.7	0.08
Bauder et al., 1978	ND	dryland	5.8	33.9	0.17
		deficient	9.7	60.2	0.16
		optimum	10.3	64.5	0.16
		excessive	10.8	68.6	0.16
		average	9.2	56.8	0.16
Retta and Hanks, 1980	UT	line source	7.2	38.3	0.19
		average	8.0	37.6	0.21
			9.3	47.7	0.20
			10.7	55.4	0.19
			11.9	57.4	0.21
			12.5	61.6	0.20
		9.9	49.6	0.20	
Sammis, 1981	NM	line source	6.0	66.7	0.09
		average	8.3	77.6	0.11
			10.0	78.5	0.13
			11.5	90.9	0.13
			12.3	92.5	0.13
			13.1	103.8	0.13
			14.3	109.4	0.13
			15.0	117.5	0.13
			17.0	124.5	0.14
			19.6	135.2	0.15
			22.1	145.7	0.15
			13.7	104.0	0.13
Carter and Sheaffer, 1983†	MN	high	7.4	32.6	0.23
		med. high	7.0	29.9	0.23
		med. low	5.5	26.4	0.21
		dryland	2.1	17.9	0.12
		average	5.5	26.7	0.21
Undersander, 1987‡	TX	line source	13.5	76.2	0.18
		average	22.6	113.0	0.20
			16.4	91.7	0.18
			17.5	93.6	0.19
Wright, 1988	ID	full irrig.	14.7	94.2	0.16
Bogler and Matches, 1990	TX	average	na§	na	0.17
Smeal et al., 1991	NM	line source	3.0	45.8	0.07
		average	6.0	63.1	0.09
			8.7	78.5	0.11
			12.3	91.3	0.13
			15.1	104.9	0.14
			14.8	106.7	0.14
			12.5	93.7	0.13
	10.3	83.4	0.12		
Overall averages					
All authors	all	all	12.8	88	0.16
All authors	all	full irrigation	16.6	91	0.19
All authors	all	deficit irrigation	11.1	80	0.17
All authors	all	dryland	6.0	39	0.14

† Harvests 3 and 4 only reported.

‡ Maximum yield and ET only reported.

§ na, not applicable.

Harvest Timing

Harvest timing has been found to influence alfalfa WUE. A study conducted in New Mexico from 1981 to 1987 by Smeal et al. (1991) evaluated the WUE response of alfalfa in relation to the accumulation of growing degree-days (G_c) within each

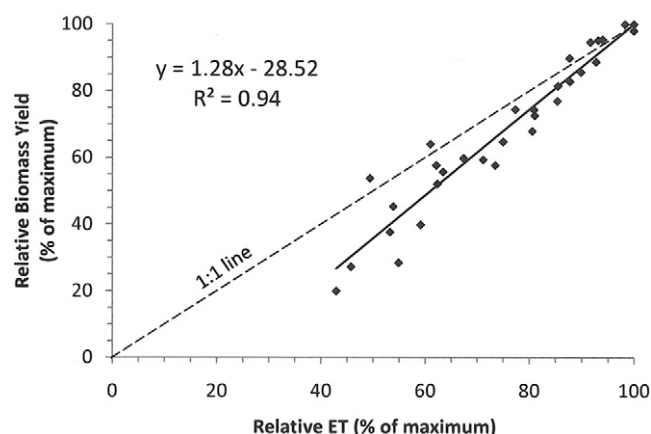


Fig. 2. A relative alfalfa water production function from studies with variable irrigation in the Great Plains and Intermountain West of the United States. The figure relates average season long evapotranspiration (ET) as a percentage of the maximum ET within the same study to biomass yield as a percent of the maximum biomass yield within the same study.

growth cycle. This study compared biomass yield per harvest to transpiration and growing degree-days with all other variables held constant. Aboveground biomass yield to transpiration ratio decreased as growing degree-day accumulation increased between the levels of 450 G_c (bud initiation) and 700 G_c (full bloom) in a very linear fashion. Smeal et al. (1991) suggested that this reduction in biomass yield per unit of transpiration may have been due to partitioning of dry matter into crowns, roots, and reproductive structures rather than new leaves and stems. This assumption was based on the previous work of Smith (1960) who evaluated establishment and management of alfalfa and Brown et al. (1972) who studied energy accumulation and utilization in plants. The accumulation of aboveground growth follows a sigmoid pattern, with growth rates declining after about the bud stage. Since the canopy has achieved full cover by this time ET has reached a maximum and continues at a near-constant level until the bloom state, while growth rates are declining. This

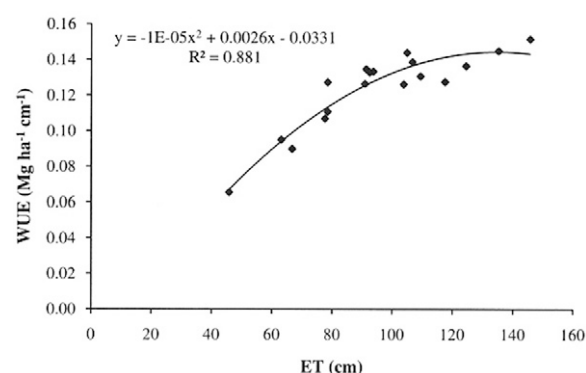


Fig. 3. Alfalfa water-use efficiency (WUE) relationship to evapotranspiration (ET) from two deficit irrigation studies in New Mexico (Sammis, 1981; Smeal et al., 1991).

suggests that the WUE of alfalfa may be increased by harvesting on a 400 to 450 G_c schedule rather than the 550 G_c schedule recommended by Selirio and Brown (1979). However, Smeal et al. (1991) also noted that harvesting alfalfa at the prebud stage (400 G_c) for any period of time will reduce plant vigor and decrease stand longevity as shown by Smith (1972) and Robinson and Massengale (1968). Growing degree-day accumulation for each harvest certainly has some impact on alfalfa WUE, but it is likely that the risk of reduced stand longevity would not justify harvesting at an early growth stage to improve WUE.

Stand Age

Stand density and vigor decline over time in alfalfa. Stands can persist for 7 to 10 yr under normal conditions with many reported exceptions. Stand density is important for production and is linked to WUE in alfalfa (Smeal et al., 1991). When comparing WUE over time, it was found that WUE was least in the establishment year and was maximized at year 5 and remained constant afterward. This may be due to a difference in CO₂ partitioning as the plant matures. Studies done by Thomas and Hill (1949) and Brown et al. (1972) found that over a 2-yr period, an

Table 2. Effect of harvest interval on alfalfa biomass yield, evapotranspiration (ET), and water-use efficiency (WUE) from studies in the Great Plains and Intermountain West of the United States. Yield, ET, and WUE were averaged over irrigation treatments and years.

Author	Location	Treatment	Harvest interval	Yield Mg ha ⁻¹	ET cm	WUE Mg ha ⁻¹ cm ⁻¹
Daigger et al., 1970	NE	full irrigation	1	5.0	54.5	0.09
			2	3.2	40.6	0.08
			3	3.3	56.6	0.06
Undersander, 1987	TX	line source	1	5.4	18.2	0.29
			2	4.5	29.9	0.15
			3	4.5	24.3	0.19
			4	3.8	19.0	0.20
			5	3.2	25.0	0.13
Wright, 1988	ID	full irrigation	1	6.7	37.5	0.18
			2	5.0	31.9	0.16
			3	3.4	28.4	0.12
Smeal et al., 1991	NM	line source	1	4.7	33.6	0.14
			2	2.7	20.9	0.13
			3	2.4	18.0	0.13
			4	1.7	19.2	0.09
			5			
Overall average	all	all	1			0.18
			2			0.13
			3			0.13
			4			0.15
			5			0.13

average of 59% of fixed CO₂ went toward shoot and leaf development in the first year as compared to 84% during the second year. It is assumed that much of the carbon that is fixed in photosynthesis in the early years is partitioned to root development for the plant. Holt et al. (1975) found that there is a time in the life of an alfalfa stand when an equilibrium is reached between photosynthates used in producing new root tissue and the death of old root tissue. Smeal et al. (1991) concluded that the amount of water transpired to produce a given amount of total above and below ground biomass remained constant over the study period, but the amount of water transpired to produce a given above ground biomass decreased until year 5 or 6 when the root system was fully developed. However, consideration of the evaporative portion of ET also affects the WUE and is subject to change as the stand ages. A relatively low WUE occurs during the establishment year due to evaporation from low ground cover. After the establishment year, full ground cover increases transpiration but reduces evaporative loss. In older, thinning stands, ground cover decreases and more water loss to evaporation can lead to reduced WUE (Nelson and Smith, 1968).

When considering adopting deficit irrigation as a water conservation approach, the implications of stand age on alfalfa WUE should be considered. Since WUE was not maximized until year 5 and remained constant afterward (Smeal et al., 1991), an alfalfa producer may improve WUE by leaving a stand of alfalfa in production for longer than the typical 4 to 5 yr production period. Lengthening the life of the stand will reduce the frequency of lower WUE found during establishment and the first few production years. However, as stands age they decline in density and are at risk for encroachment by weeds. Therefore, a balance between increasing WUE and a thinning stand must be found.

Water Table Contribution to Alfalfa Evapotranspiration

Alfalfa has an extensive root system with the ability to extract water from deep within the soil profile. Alfalfa roots commonly grow 2 to 4 m in length and under the right conditions, can grow as deep as 9 m (Peterson, 1972). In the presence of a shallow water table, the contribution to ET from the water table can be significant. Documenting water uptake from shallow groundwater may be important if controlled deficit or partial season irrigation of alfalfa is promoted as a water savings approach (Putnam et al., 2005). Additionally, water uptake from ground water may have an effect on nearby streamflows which can impact other water users.

Research that quantifies the water table contribution to alfalfa ET is very limited, especially under conditions of deficit irrigation. A study in North Dakota performed over 2 yr by Benz et al. (1984) used nonweighing lysimeters under three irrigation levels (0.3, 0.8, and 1.3 times potential ET) and four water table depths (46, 101, 155, and 210 cm) on sandy loam and loamy sand soils. The ET was affected by both water table depth and irrigation level. Contribution to ET from the water table ranged from 0% for the 155 and 210 cm deep water table conditions to 57% for the 46 cm water table depth at the 0.3 irrigation level.

Another study completed over 4 yr on the western slope of Colorado by Kruse et al. (1993) used weighing lysimeters on a fine sandy loam soil to evaluate the effects of three water table depths (no water table, 60 cm, and 105 cm) and two levels of water salinity (0.65 ds m⁻¹ to simulate river water and 6 ds m⁻¹

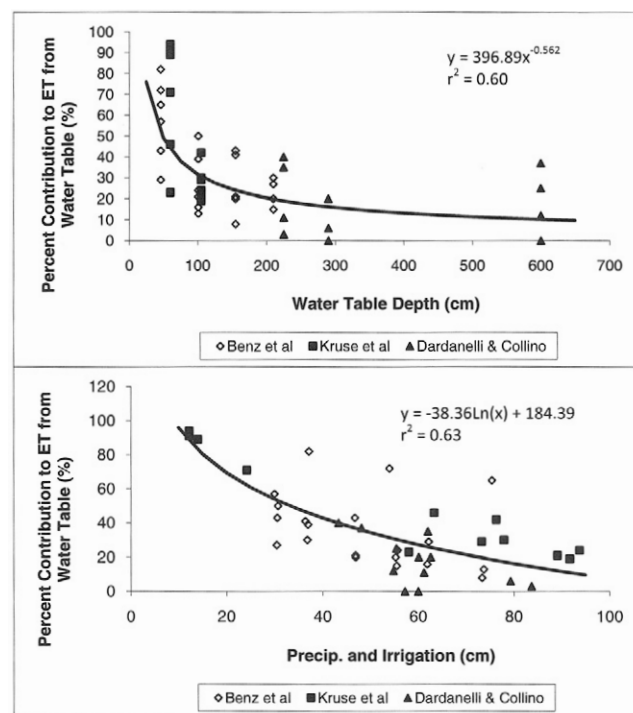


Fig. 4. Relationship between percent contribution to evapotranspiration (ET) from the water table and water-table depth (upper) and water from precipitation and irrigation for alfalfa (lower).

to simulate saline groundwater). Over the course of the study at the 60 cm water table depths, water from the water table made up 62 and 76% of alfalfa seasonal ET for the 0.65 and 6 ds m⁻¹ water, respectively. At the 105 cm water table depths, the water table supplied 27 and 28% of the alfalfa's seasonal ET for the 0.65 and 6 ds m⁻¹ water, respectively.

Dardanelli and Collino (2002) conducted a study in the Argentine Pampas to determine the water table contribution to alfalfa water use. The study took place on four different sites, one where alfalfa did not have access to a water table and three with water tables varying from 2.0 to 4.0 m. The study used a water balance approach from the site without a water table to estimate ET and then used measured biomass yield to determine the local WUE. At the sites with a water table, water table contributions to ET were estimated by dividing the dry matter biomass yield by the local WUE then subtracting the change in soil water for the top 1.2 m. The average contribution to ET from the water table was 21% at the lower depths. This study found that water table contribution to ET increased as current seasonal rainfall decreased.

While additional research is needed to understand the contribution to ET from shallow groundwater, a relationship between ET contribution from the water table and depth to the water table was estimated for alfalfa based on results of published data (Fig. 4). The ET contribution declines rapidly as water table depth increases to about 100 cm and then levels off to about 20% for depths up to 600 cm. There is also a strong negative relationship between contribution to ET from the water table and the amount of irrigation and precipitation (Fig. 4), suggesting that the potential of significant ET contribution from a water table would increase under controlled deficit or partial season irrigation. Evapotranspiration determined on the basis of applied irrigation will underestimate actual crop

ET where a deficit irrigation system is being considered and shallow water table conditions exist.

SUMMARY AND CONCLUSIONS

Data from several studies conducted across the Great Plains and Intermountain West of the United States were examined to evaluate the biomass yield response of irrigated alfalfa to ET and to determine how that response is affected by deficit irrigation strategies (Daigger et al., 1970; Bauder et al., 1978; Retta and Hanks, 1980; Sammis, 1981; Carter and Sheaffer, 1983; Undersander, 1987; Wright, 1988; Bogler and Matches, 1990; Smeal et al., 1991). Only studies that accounted for water from all sources, including soil water, were reviewed to elucidate management practices that can improve WUE in deficit irrigated alfalfa. The results of these studies revealed a positive, linear response of alfalfa biomass yield to ET and suggested several management techniques that could improve that relationship. Total seasonal ET of alfalfa ranged from 18 cm under dryland to 152 cm under full irrigation with an average total seasonal ET of 88 cm. The irrigation strategy used to reduce ET in most reported studies was deficit irrigation, with a reduced amount of irrigation water applied throughout the entire crop growing season. A relative water production function accounted for most of the environmental and climate variability among studies and showed that when managing irrigation to reduce ET, yield declines from its local maximum 25% faster than ET.

For environmental conditions in the Great Plains and Intermountain West, alfalfa WUE is greatest during the first harvest interval when plants are using carbohydrates stored in roots, solar irradiance is high, and temperatures are relatively low. Thus, partial season irrigation is a potential management approach to saving irrigation water that combines full irrigation during the earliest growth cycle with no irrigation during subsequent growth cycles. Irrigation termination during hot, dry summers can be detrimental to alfalfa stands in arid environments and sandy soils (Ottman et al., 1996), but evidence suggests minimal stand loss in semiarid environments with finer soil textures (Hansen and Putnam, 2000). Partial season irrigation may improve total seasonal WUE for alfalfa compared to deficit irrigation and may be a better approach for conservation of agricultural water compared to deficit irrigation or conversion to dryland production.

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